

Finding a percentage increase or decrease

One multiplier does the whole job — and the reverse question needs a division.

LESSON 63–66

4 × 40 MIN

MATEMATIKA 6, §14

S7 10

LESSON CLOCK

30'

40'

50'

30'

10

The multiplier	30 min
Finding the change	40 min
The reverse question	50 min
Profit, loss and discount	30 min
Homework	10 min

LEARNING OBJECTIVES

- Increase and decrease a quantity by a percentage.
- Find the percentage change from the old and new values.
- Find the original value after a change.
- Explain why a rise and a fall of the same percentage do not cancel.

TEXTBOOK

National	pp. 179–190
Cambridge	Stage 7, pp. 96–104
Workbook	Exercise 10.3

01

Explanation

The multiplier

new = old × (1 ± $\frac{P}{100}$)

CHANGE	MULTIPLIER	80 000	250
up 20%	1.2	96 000	300
down 20%	0.8	64 000	200
up 7%	1.07	85 600	267.5
down 35%	0.65	52 000	162.5

ONE STEP, NOT TWO

Finding 20% and then adding it works, but · 1.2 is one keystroke and cannot lose the original amount along the way.

Finding the change

$$\text{percentage change} = \frac{\text{new} - \text{old}}{\text{old}} \times 100\%$$

OLD	NEW	CHANGE	PERCENTAGE
40	50	+10	+25%
50	40	−10	−20%
80 000	92 000	+12 000	+15%
200	150	−50	−25%

ALWAYS DIVIDE BY THE OLD VALUE

The first two rows use the same pair of numbers and give different percentages, because the base changes. A rise of 25% is undone by a fall of 20%, not of 25%.

The reverse question

“After a rise of 20% the price is 96 000” — the 96 000 is the **new** value, so divide by the multiplier.

GIVEN	WRONG	RIGHT	ANSWER
after +20% it is 96 000	96 000 − 19 200	96 000 ÷ 1.2	80 000
after −20% it is 64 000	64 000 + 12 800	64 000 ÷ 0.8	80 000
after −25% it is 150	150 + 37.5	150 ÷ 0.75	200

CHECK BY GOING FORWARDS

80 000 · 1.2 = 96 000 ✓. The forward check takes five seconds and settles whether the division was the right move.

Profit, loss and discount

SITUATION	WORKING	ANSWER
bought at 40 000, sold at 50 000	$\frac{10\,000}{40\,000} \cdot 100$	25% profit
bought at 50 000, sold at 40 000	$\frac{10\,000}{50\,000} \cdot 100$	20% loss
a coat at 300 000 with 30% off	$\cdot 0.7$	210 000
a bill of 120 000 plus 12% tax	$\cdot 1.12$	134 400

PROFIT AND LOSS ARE PERCENTAGES OF THE COST

A trader who buys at 40 000 and sells at 50 000 makes 25%, not 20% — the cost price is the base. Using the selling price instead is the classic error.

02

Worked examples

EXAMPLE 1 Increase 80 000 by 15%.

1

Multiplier: 1.15.

2

$80\,000 \cdot 1.15$

3

$= 92\,000$

ANSWER

92 000

EXAMPLE 2 A price falls from 200 to 150. Find the percentage decrease.

① Change: -50 .

② $\frac{50}{200} \cdot 100$

The old value underneath.

③ $= 25\%$ decrease.

ANSWER

25%

EXAMPLE 3 After a fall of 20% a price is 64 000. What was it before?

① The multiplier was 0.8.

② $64\ 000 \div 0.8$

Divide, do not add.

③ $= 80\ 000$

Check: $80\ 000 \cdot 0.8 = 64\ 000$ ✓

ANSWER

80 000

03 Interactive model

Ask whether a shop that raises a price 20% and then takes 20% off has returned to the start; the near-unanimous “yes” is worth correcting in public.

Multiplier, change, or reverse?

The multiplier for a rise of 15%:

The multiplier for a fall of 35%:

80 000 up 15%:

From 40 to 50:

+20%

+25%

+10%

+50%

From 50 to 40:

-20%

-25%

-10%

-80%

After +20% a price is 96 000; before:

76 800

80 000

115 200

86 400

Bought at 40 000, sold at 50 000:

20% profit

25% profit

10% profit

25% loss

Up 20% then down 20% gives:

the same price

96% of it

104% of it

90% of it

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Increase	Ortish	Увеличение
Decrease	Kamayish	Уменьшение
Multiplier	Ko'paytuvchi	Множитель
Percentage change	Foizli o'zgarish	Процентное изменение
Original value	Boshlang'ich qiymat	Первоначальное значение
Discount	Chegirma	Скидка
Profit	Foyda	Прибыль
Loss	Zarar	Убыток

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

To increase by 7% multiply by:

To decrease by 35% multiply by:

Percentage change divides by:

After -25% a price is 150; before:

187.5

200

175

112.5

A rise of 25% is undone by a fall of:

25%

20%

30%

75%

Profit is a percentage of:

the selling price

the cost price

the profit

the tax

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 80 000 up 20%

ANSWER 96 000

2. 80 000 down 20%

ANSWER 64 000

3. 250 up 7%

ANSWER 267.5

4. 250 down 35%

ANSWER 162.5

5. The multiplier for +15%

ANSWER 1.15

6. The multiplier for -35%

ANSWER 0.65

7. 300 000 with 30% off

ANSWER 210 000

Medium · the standard question

7 PROBLEMS

1. From 40 to 50

ANSWER +25%

2. From 50 to 40

ANSWER -20%

3. From 80 000 to 92 000

ANSWER +15%

4. From 200 to 150

ANSWER -25%

5. After +20% it is 96 000: before

ANSWER 80 000

6. After -20% it is 64 000: before

ANSWER 80 000

7. A bill of 120 000 plus 12% tax

ANSWER 134 400

Hard · stretch and reason

7 PROBLEMS

1. Bought at 40 000, sold at 50 000: the profit percentage

ANSWER 25%

2. Bought at 50 000, sold at 40 000: the loss percentage

ANSWER 20%

3. Up 20% then down 20% from 500

ANSWER 480

4. A price after two rises of 10% from 200

ANSWER 242

5. After -40% a price is 72 000: before

ANSWER 120 000

6. A coat costs 210 000 after 30% off: the original

ANSWER 300 000

7. Why is a 25% rise undone by a 20% fall?

ANSWER The fall is taken from a larger amount

07 Homework

Homework — 5 tasks

Every reverse question is a division; check it by multiplying forwards.

1. Increase 45 000 by 16% and decrease it by 16%.
2. A price rises from 250 to 310. Find the percentage increase.
3. After a fall of 25% a price is 180 000. Find the original price.
4. A trader buys at 60 000 and sells at 75 000. Find the profit percentage.
5. Show that a rise of 50% followed by a fall of 50% leaves 75% of the original.